Examiner only

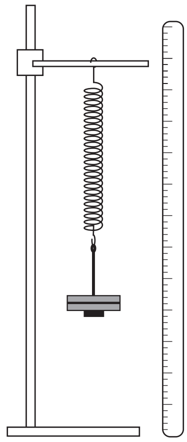
7. A Welsh company has been asked to design a toy that is able to launch a model plastic plane. They decide to use a spring mechanism in its design. Initially they carry out an experiment to investigate whether a spring is suitable for use in the toy.

Slotted masses are added to exert a force on the spring.

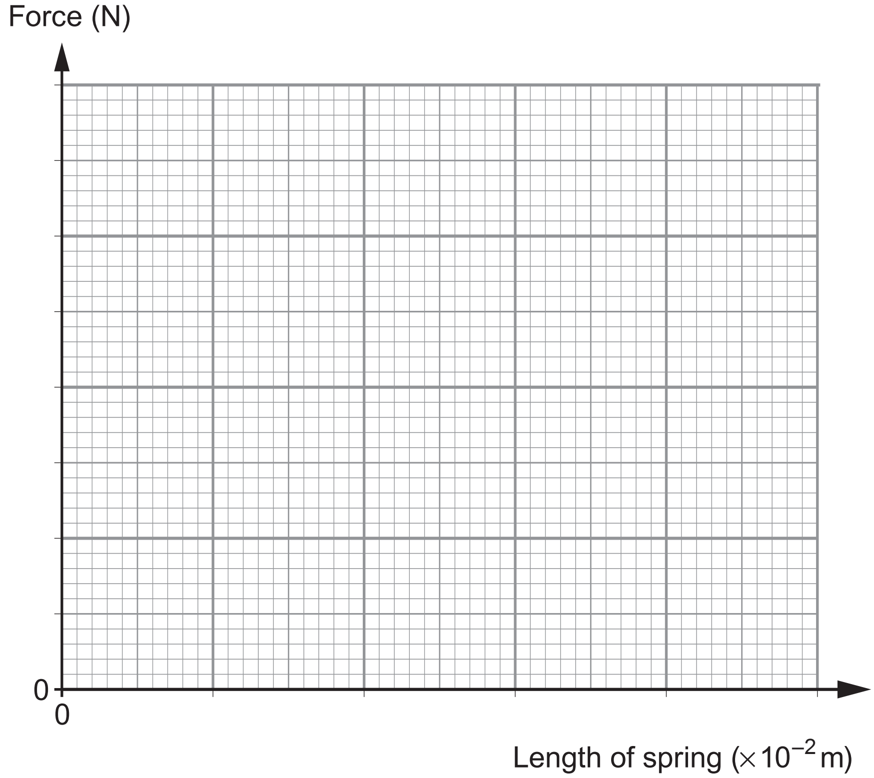
The resulting length of the spring is measured.

The results are shown in the table below.

Mass (g)	Force (N)	Length of spring ($\times 10^{-2}$ m)
100	1.0	7.5
200	2.0	10.0
300	3.0	12.5
600	6.0	20.0
700	7.0	22.5



(a) (i) Plot the data on the grid below and draw a suitable line. [4]



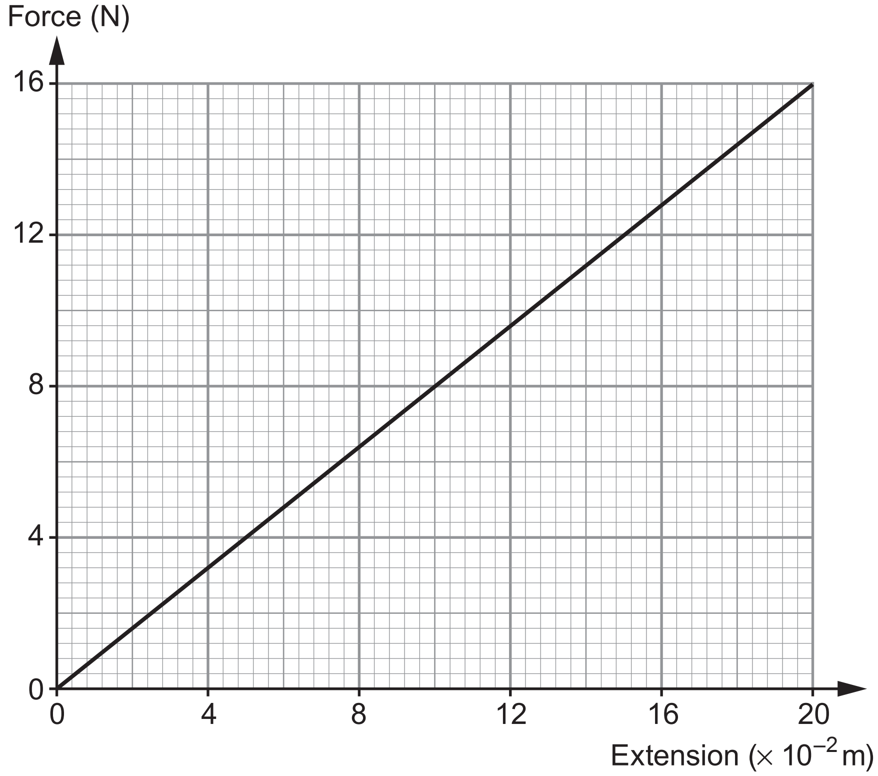
(ii) State the length of the spring when no force is applied. [1]

Length of spring = ... $\times 10^{-2}$ m

Examiner only

(b) The design company decides to connect two of these springs in parallel with each other in the launch mechanism of the model plastic plane.

The graph shows how the extension of this spring system changes with force.



The two spring system is stretched and a model plastic plane of mass 55×10^{-3} kg is attached. A trigger is pressed, which releases one end of the spring system, and the plane is launched.

(i) The two spring system is stretched to a maximum extension of 15×10^{-2} m. Use the graph and an equation from page 2 to calculate the work done stretching this spring system. Give an appropriate unit with your answer. [4]

Work done = ...

Unit ....

TURN OVER FOR THE REST OF THE QUESTION.

Examiner only

(ii) Health and safety guidelines limit the launch velocity of the plastic plane to a maximum of 4 m/s. Using an equation from page 2 and your answer from *(b)*(i), calculate the launch velocity of this plane (mass = 55×10^{-3} kg) **and state** whether the launch system design meets this safety guideline. [3]

Launch velocity = ... m/s

....

....

(iii) Explain why the actual launch velocity is always less than your calculated launch velocity. [2]

....

....

....

(iv) A student suggests that a maximum momentum value would be more appropriate than stating a maximum velocity restriction for the launch. Explain why this would be an improvement to the health and safety guidelines. [2]

....

....

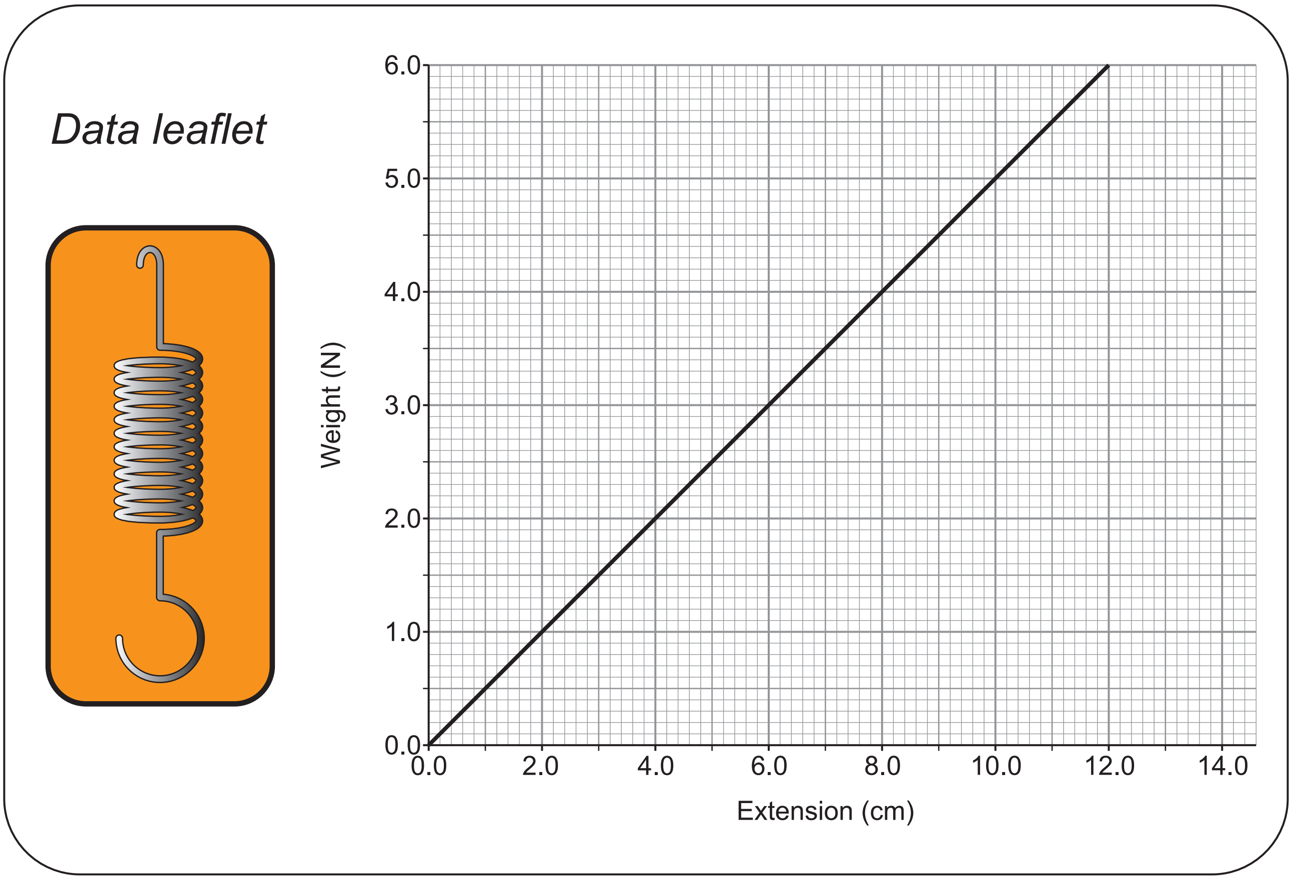
....

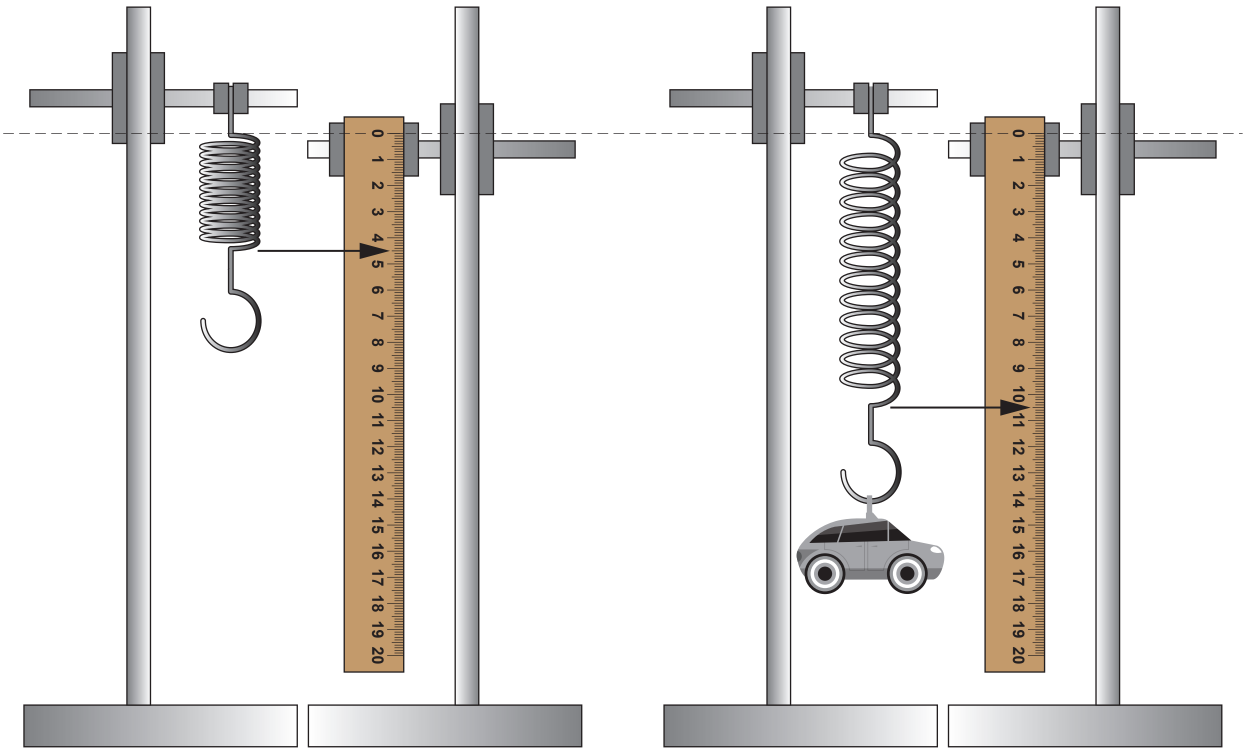
....

16

END OF PAPER

8. A school science technician bought a spring for an experiment.
The data leaflet about the spring is shown below.





Picture 1

Picture 2

Examiner only

(a) A student used the spring to determine the mass of a toy.

Two pictures of the experiment were taken by the student.
Picture 1 shows the unstretched spring and Picture 2 shows the spring with the toy attached.

(i) Use information from the two pictures to determine the spring's extension with the toy attached. [1]

spring's extension = ... cm

(ii) Use information from the leaflet, and an equation from page 2, to calculate the mass of the toy. (g = 10 N/kg) [3]

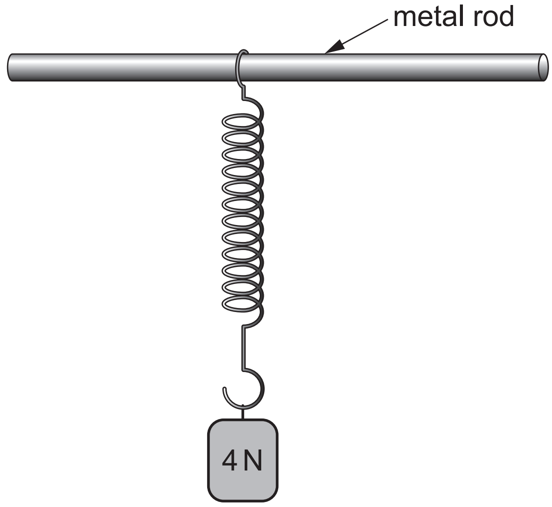
mass of toy = ... kg

(b) (i) State Newton's 3rd law. [2]

....

....

(ii) The diagram below shows the spring suspended from a metal rod.
A weight of 4 N is attached to the spring.

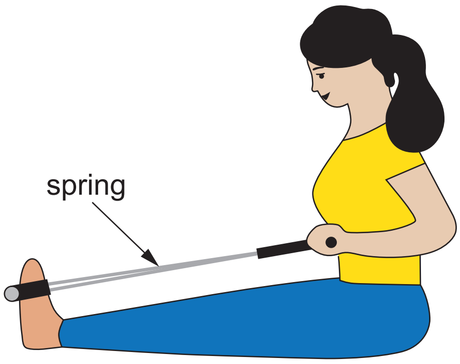


State the **size and direction** of the force that the spring applies to the weight. [2]

size = ... N   direction = ...

Examiner only

(c) The picture shows a woman using a piece of gym equipment containing a **different** spring.



Use the equation:

$$W = \frac{1}{2}Fx$$

to answer the following questions.

(i) Initially the spring is unstretched.
The woman applies a force of 35 N and the spring extends 0.32 m.
Calculate the work done by the woman stretching the spring. [2]

work done = ... J

(ii) A gym instructor suggests to the woman, if she applies **double** the force to the spring she will double the amount of work done.
Assuming the spring obeys Hooke's law (applied force is directly proportional to extension) determine whether the gym instructor is correct. [3]
Space for calculation.

....

....

....

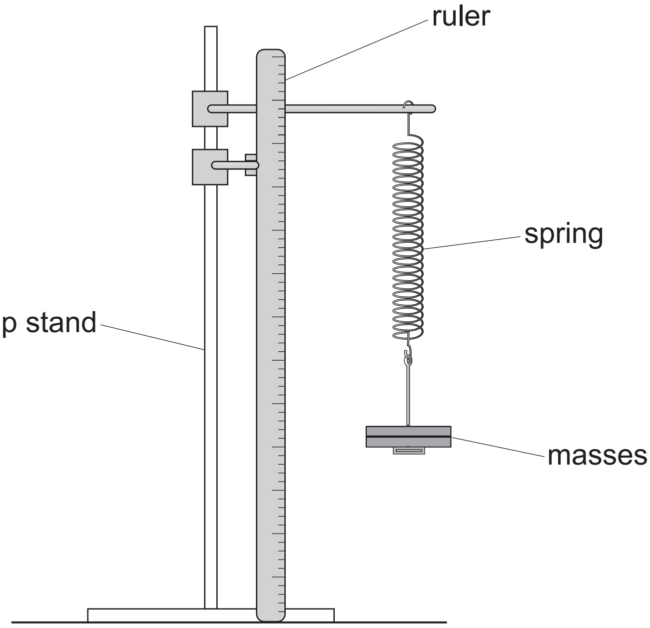
....

END OF PAPER

13

Examiner only

5. Some students carry out an experiment to determine the relationship between the force applied to a spring and its extension.
They use the following apparatus.



They collect data by loading and then unloading the spring. Their results are shown in the table below.

Mass (g)	Force (N)	Extension (cm)	
		Loading	Unloading
0	0	0.0	0.0
100	1	4.4	4.2
200	2	8.9	9.3
300	3	13.6	13.7
400	4	18.2	18.5
500	5	22.5	22.5

(a) The uncertainty in extension is given by:

$$\text{uncertainty in extension} = \frac{\text{maximum value of extension} - \text{minimum value of extension}}{2}$$

(i) State the mass with the greatest uncertainty in extension. [1]

value of mass = ... g

 (3420UB0-1)

Examiner only

(ii) Calculate this uncertainty. [2]

uncertainty = ... cm

(b) Suggest how the students could adapt the apparatus to reduce the uncertainty in their results. [1]

....

(c) Explain how they could use **all** of the results to determine a mean value for the spring constant. [3]

....

....

....

....

....

(d) The students determine the spring constant to be 22.3 N/m.
The true value of the spring constant is 22.2 N/m.
Explain what this tells the students about their data. [2]

....

....

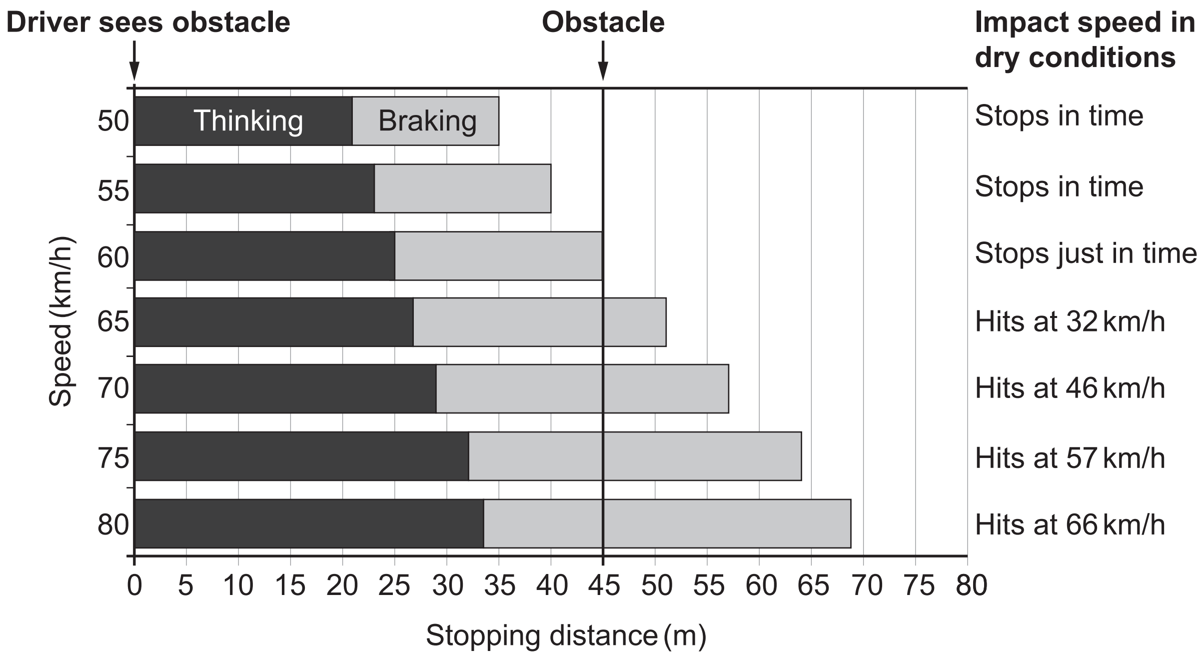
....

9

Examiner only

7. The chart below is used by traffic collision investigators. It gives the thinking, braking and stopping distances of cars driven at different speeds by an alert driver on a dry road.

An alert driver notices an obstacle 45 m away on the road ahead. The position of this obstacle is represented by the dark vertical line. If there is a collision, the chart also shows the impact speed with the obstacle.



(a) State how the following information in the chart for a speed of 70 km/h would compare if the tyre treads on the car are worn below the legal limit. [3]

(i) Thinking distance

....

(ii) Braking distance

....

(iii) Impact speed

....

Examiner only

(b) Use the information opposite to answer the following questions about a car travelling at 60 km/h which **decelerates to a stop**.
(10 km/h = 2.8 m/s)

(i) **Complete** the following table. [4]

Initial speed (km/h)	Initial speed (m/s)	Thinking distance (m)	Braking distance (m)	Stopping distance (m)
60	....	....	....	....

(ii) Using the information in the table, calculate the **thinking time** of the driver. [3]

thinking time = ... s

(iii) Use the equation:

$$v^2 = u^2 + 2ax$$

and the information in the table to calculate the deceleration of the car. [3]

deceleration = ... m/s^2

(c) The chart shows that a car travelling faster than 60 km/h will not stop in time and therefore collides with the obstacle.

(i) Explain, in terms of Newton's 1st Law, how seat belts provide protection during a collision. [2]

....

....

....

Examiner only

(ii) Explain, in terms of Newton's 3rd law, why the car and the obstacle are damaged during a collision. [2]

....

....

....

(iii) Explain, in terms of Newton's 2nd law, why cars have crumple zones. [2]

....

....

....

(d) The Royal Society for the Prevention of Accidents (RoSPA) says that "The majority of pedestrian casualties (in the UK) occur in built-up areas: 22 of the 26 child pedestrians and 264 of the 372 adult pedestrians who were killed in 2017, died on roads in built-up areas."
They strongly feel that reducing the speed limit in built-up areas from 50 km/h to 35 km/h would greatly improve these accident statistics.
Explain whether you agree with RoSPA. [3]

....

....

....

....

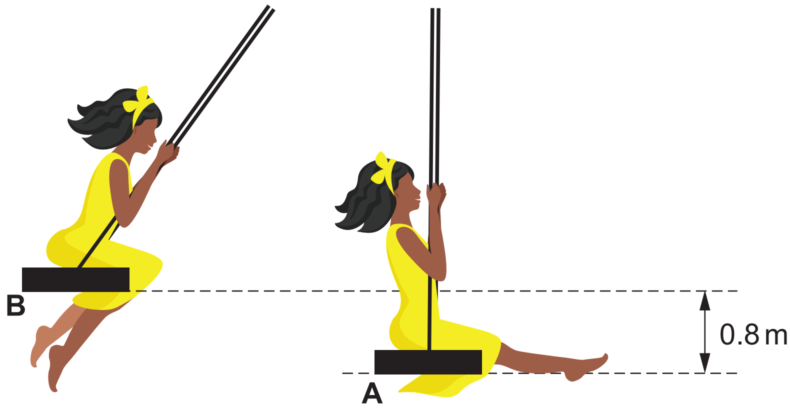
....

22

END OF PAPER

Examiner only

4. A child of mass 35 kg is playing on a swing.



The swing is pulled back to position **B**, 0.8 m above the start point, and released.

(a) (i) Use an equation from page 2 to calculate the potential energy gained by the child when moving from position **A** to position **B**. [2]
[Gravitational field strength, g = 10 N/kg]

potential energy = ... J

(ii) State the kinetic energy of the child as they pass through point **A**. Assume there are no resistive forces acting. [1]

kinetic energy = ... J

(iii) Use an equation from page 2 to calculate the velocity of the child as they pass through point **A**. [3]

velocity = ... m/s

Examiner only

(b) (i) At position **B** the child drops a ball which hits the ground time t = 0.50 s later.
The ball has an initial velocity, u = 0 m/s.
The downwards acceleration, $a = g$ = 10 m/s^2.

Use an equation from page 2 to determine the distance, x, that the ball falls. [3]

x = ... m

(ii) After the collision with the ground, the ball rolls away with initial kinetic energy of 6.4 J and then comes to a halt in 3.0 m.

Use an equation from page 2 to calculate the size of the friction force between the ball and the ground. [2]

friction force = ... N

11